

Fake News Detection Using Machine Learning



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Agenda

- ▶ Introduction
- ▶ Problem Statement
- ▶ Objectives
- ▶ Dataset
- ▶ Machine Learning Process
- ▶ Algorithms
- ▶ Results
- ▶ Future Scope



Introduction

- ▶ Fake news spreads rapidly through social media.
- ▶ Manual verification is difficult.
- ▶ Machine Learning can classify news automatically.
- ▶ NLP helps analyze textual content.



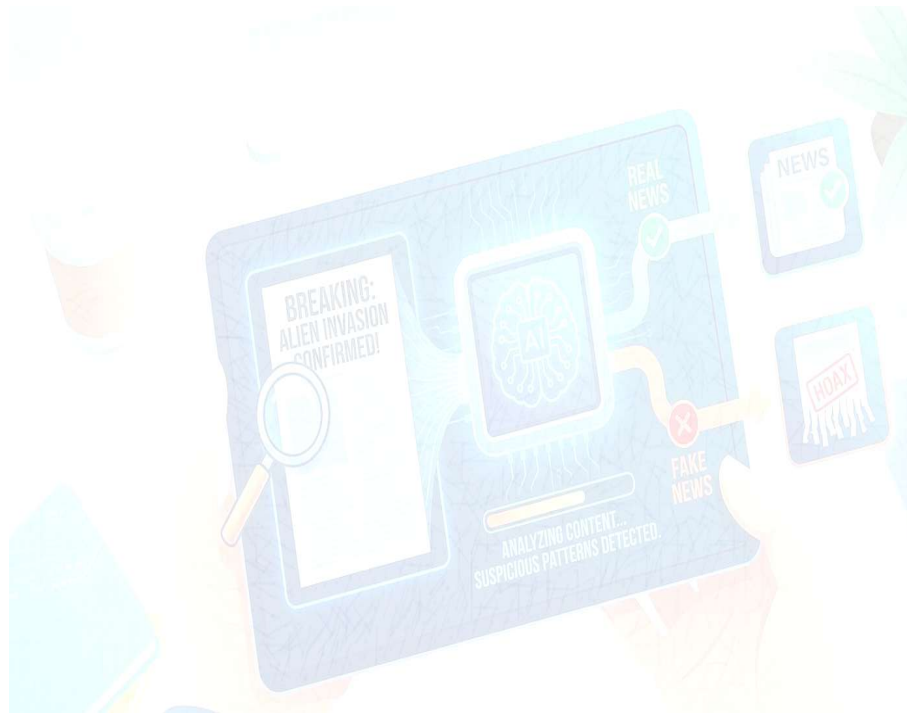
Problem Statement

- ▶ Increasing misinformation online.
- ▶ Difficult to verify every article manually.
- ▶ Need for automated fake news detection.



Objectives

- ▶ Detect fake news accurately.
- ▶ Reduce misinformation.
- ▶ Improve decision making.
- ▶ Build a reliable ML model.



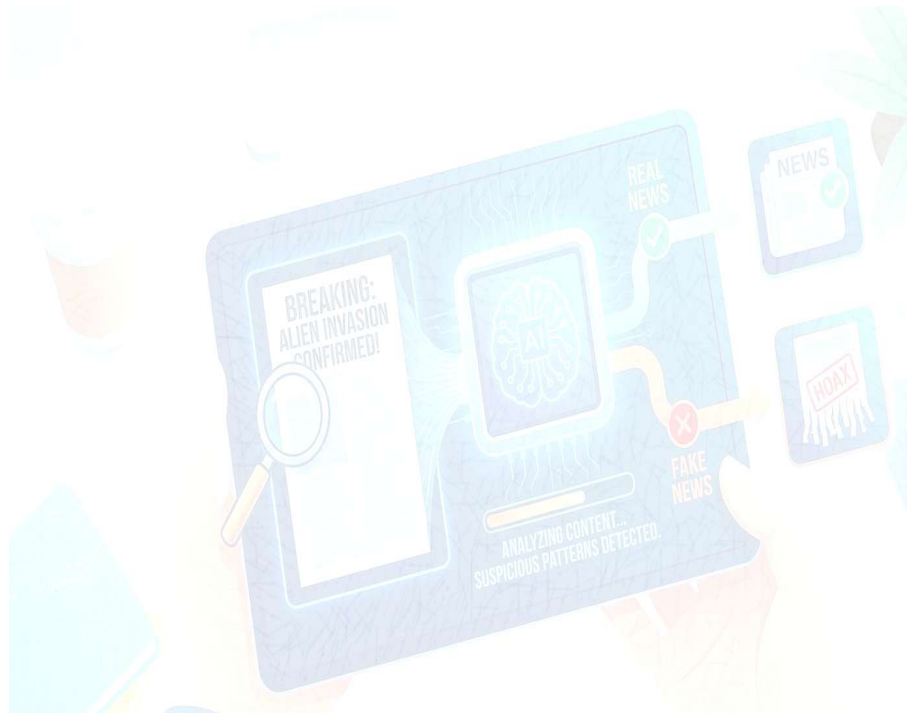
Dataset

► Dataset Files:

- • Fake.csv
- • True.csv

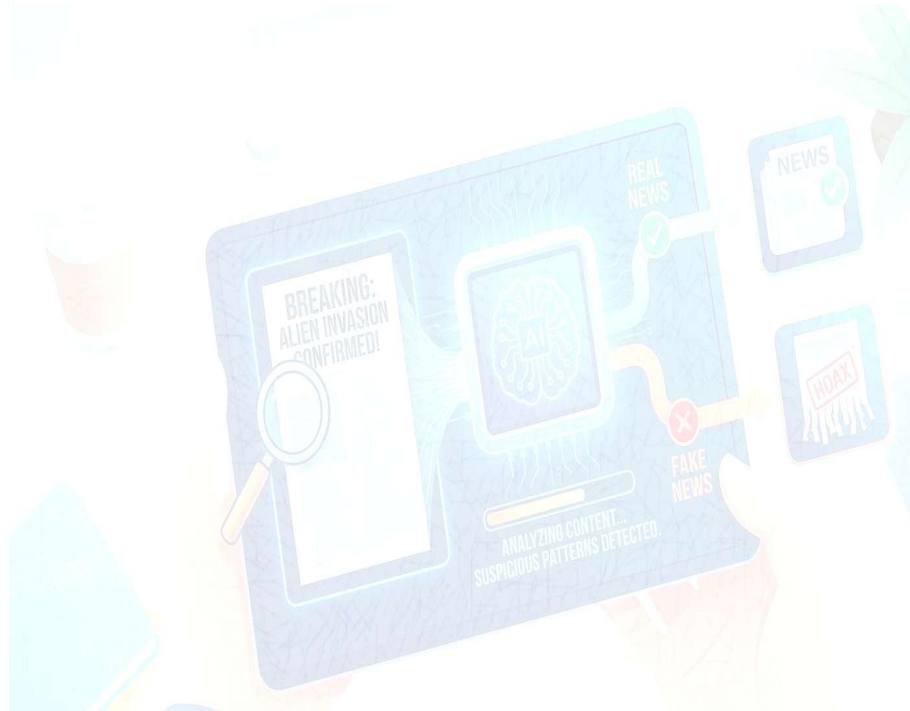
► Features:

- • Title
- • Text
- • Subject
- • Date



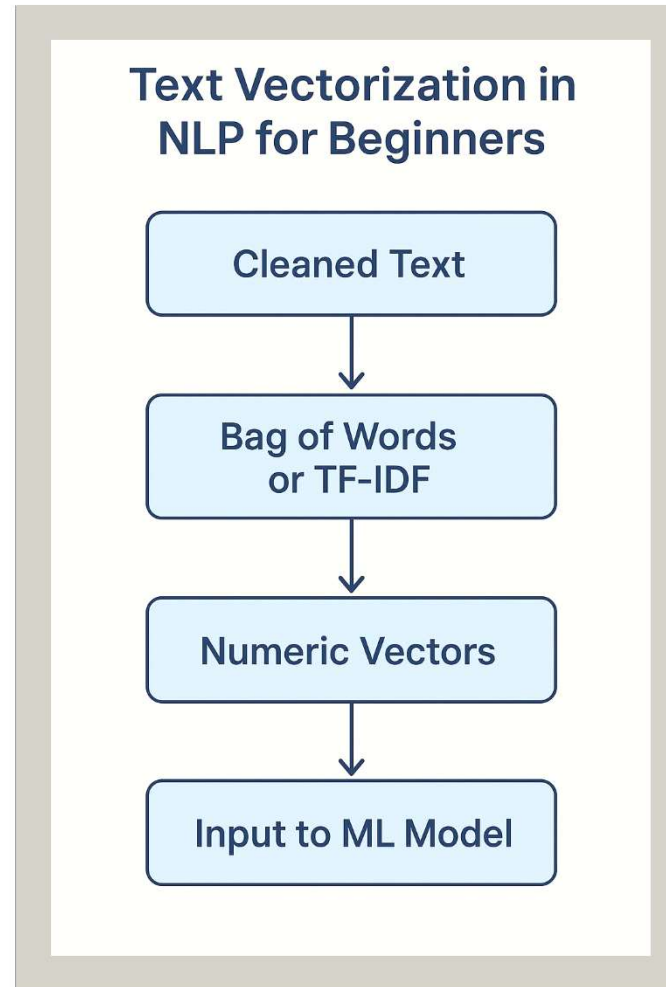
Data Preprocessing

- ▶ Remove Null Values
- ▶ Remove Stop Words
- ▶ Tokenization
- ▶ Stemming
- ▶ Lowercase Conversion



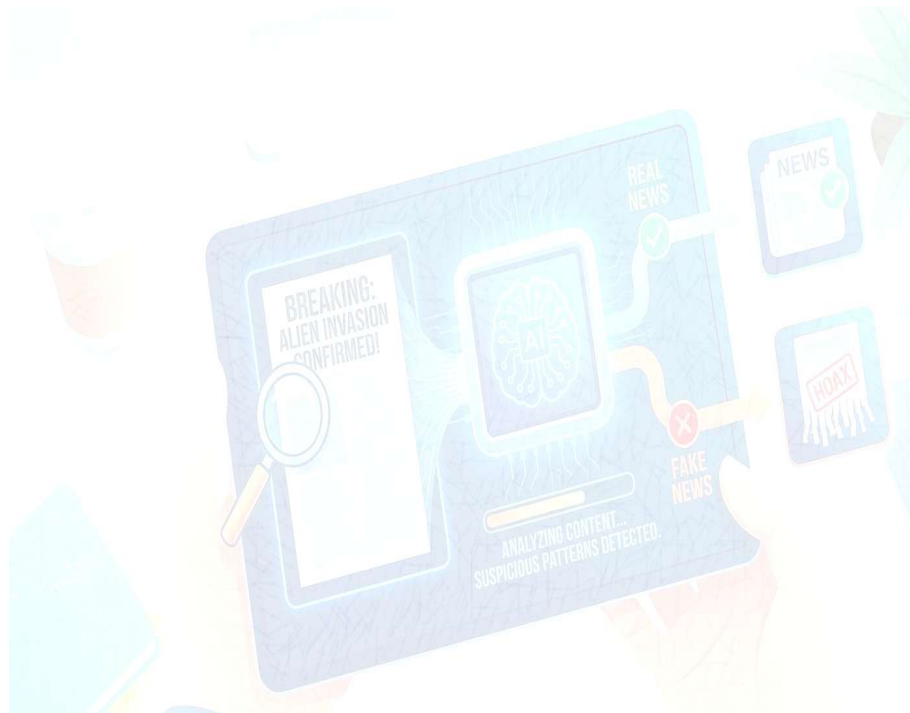
Feature Extraction

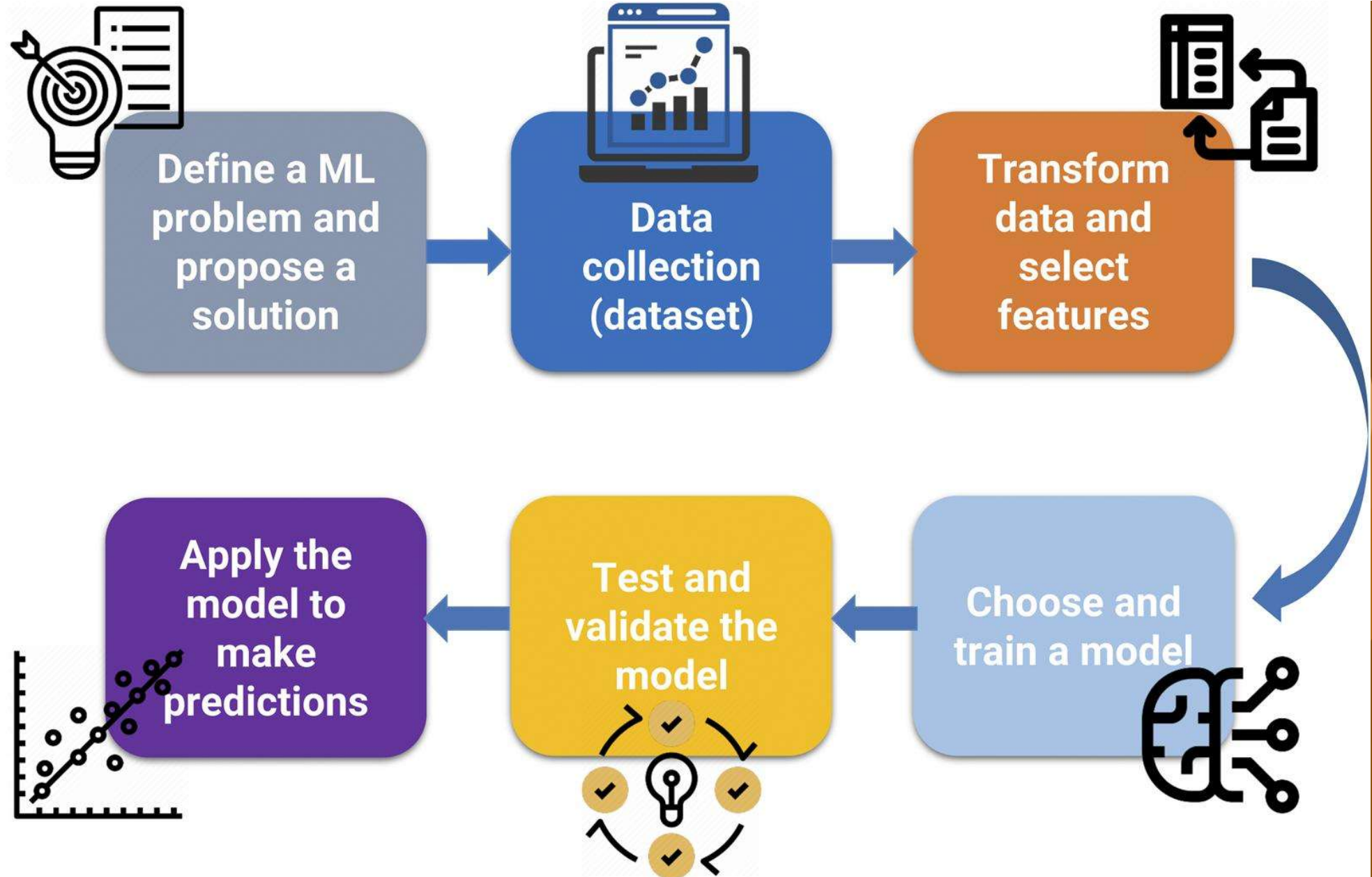
- ▶ TF-IDF Vectorizer
- ▶ Count Vectorizer
- ▶ Convert Text into Numerical Features



Machine Learning Algorithms

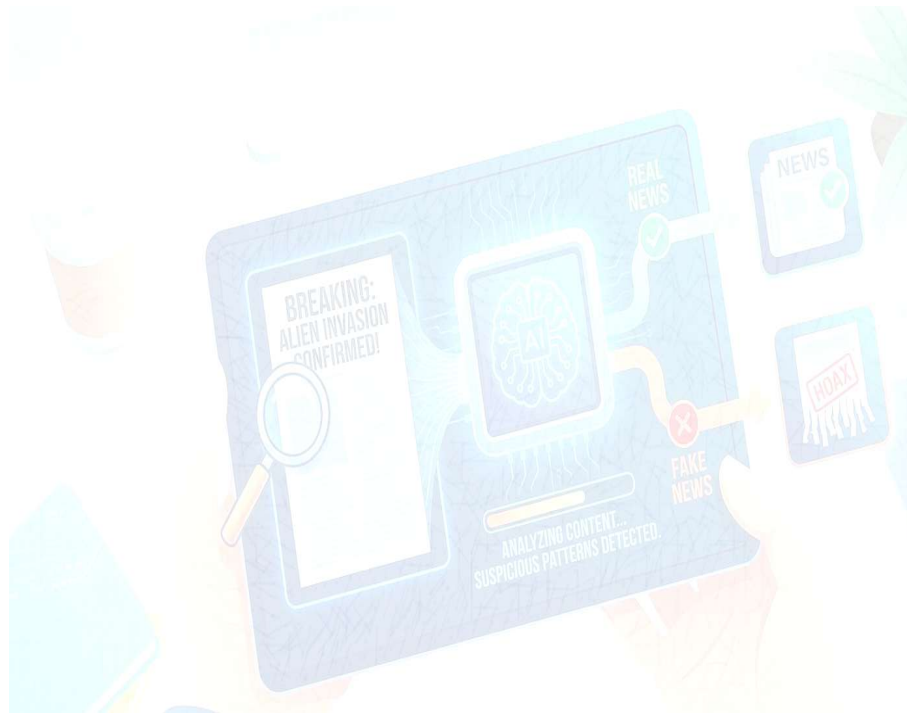
- ▶ Logistic Regression
- ▶ Decision Tree
- ▶ Random Forest
- ▶ Gradient Boosting





Workflow

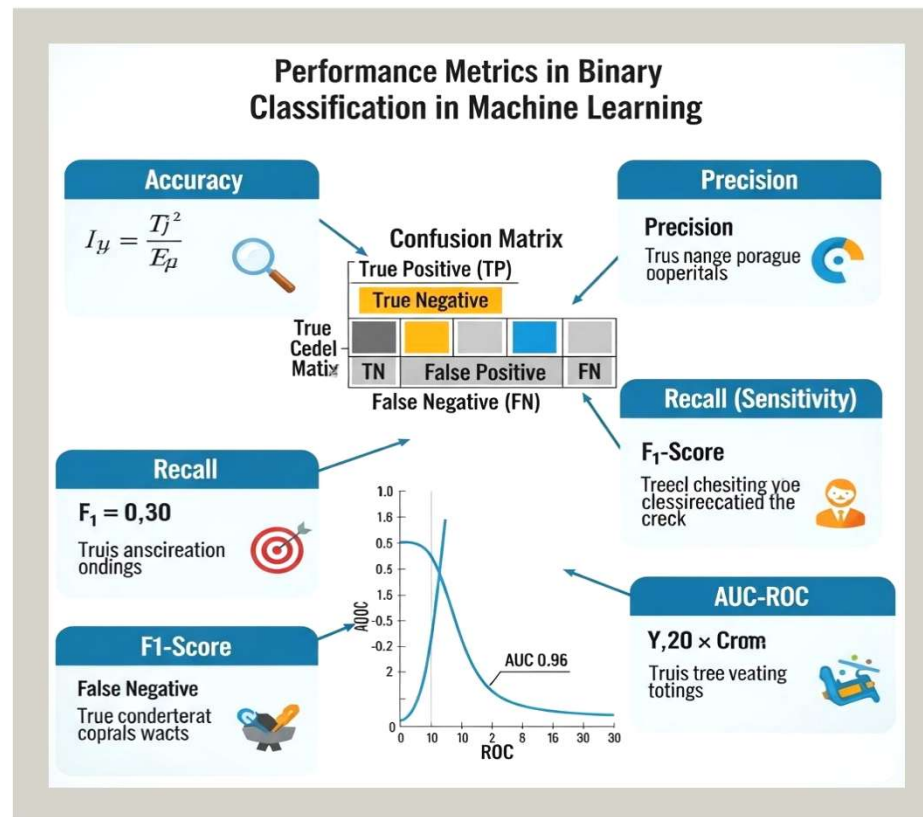
- ▶ Input Dataset
- ▶ Preprocessing
- ▶ Feature Extraction
- ▶ Train ML Model
- ▶ Prediction



Model Evaluation

► Evaluation Metrics:

- • Accuracy
- • Precision
- • Recall
- • F1-Score



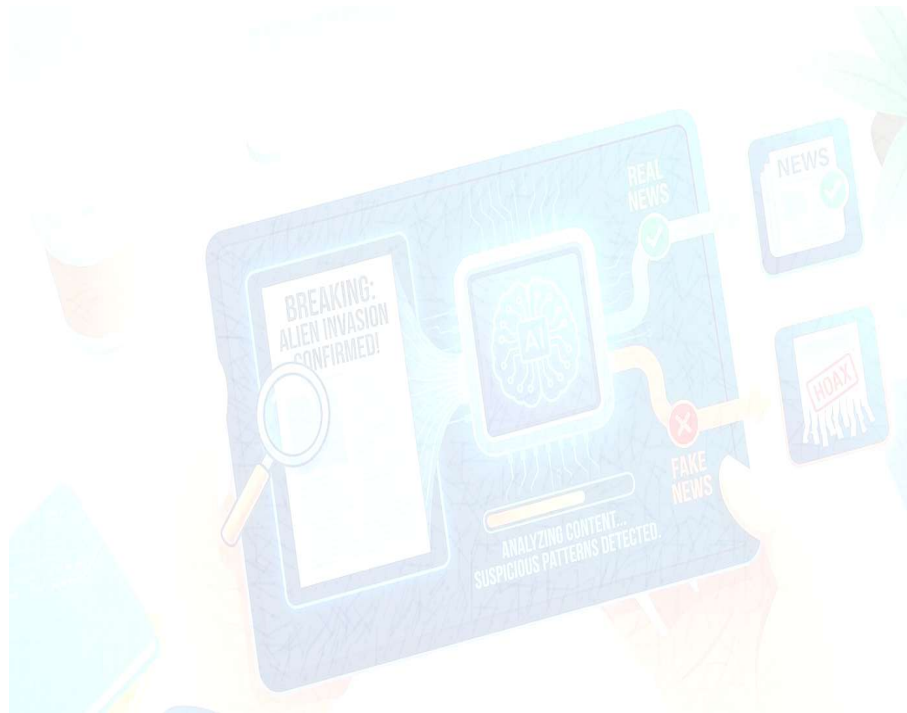
Results

- ▶ **Example Accuracy:**
- ▶ Logistic Regression - 98%
- ▶ Decision Tree - 99%
- ▶ Random Forest - 98%
- ▶ Gradient Boosting - 99%



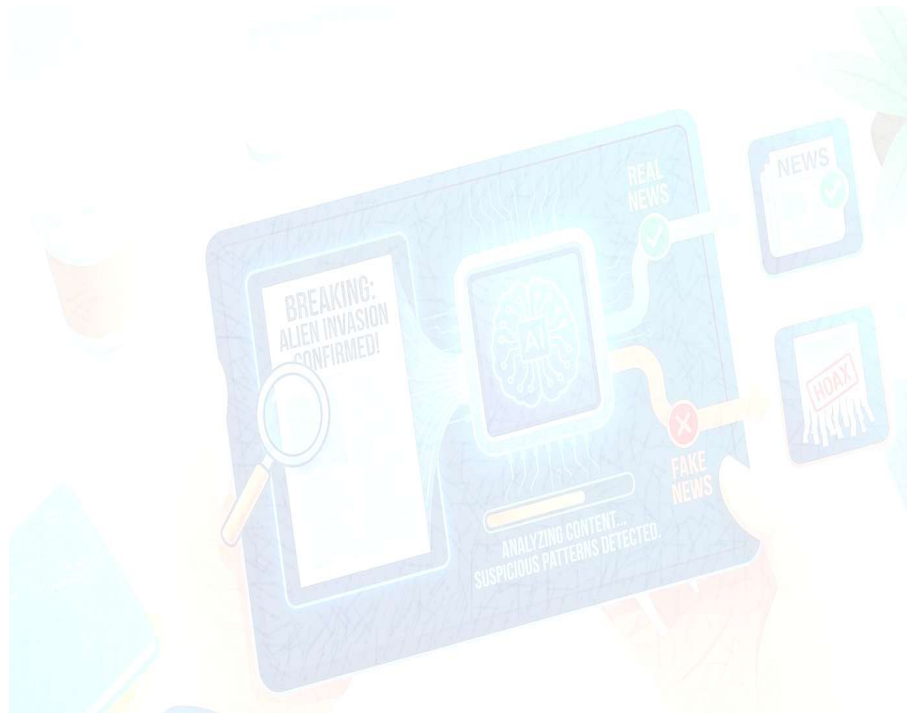
Advantages

- ▶ Fast Prediction
- ▶ High Accuracy
- ▶ Reduces Human Effort
- ▶ Scalable



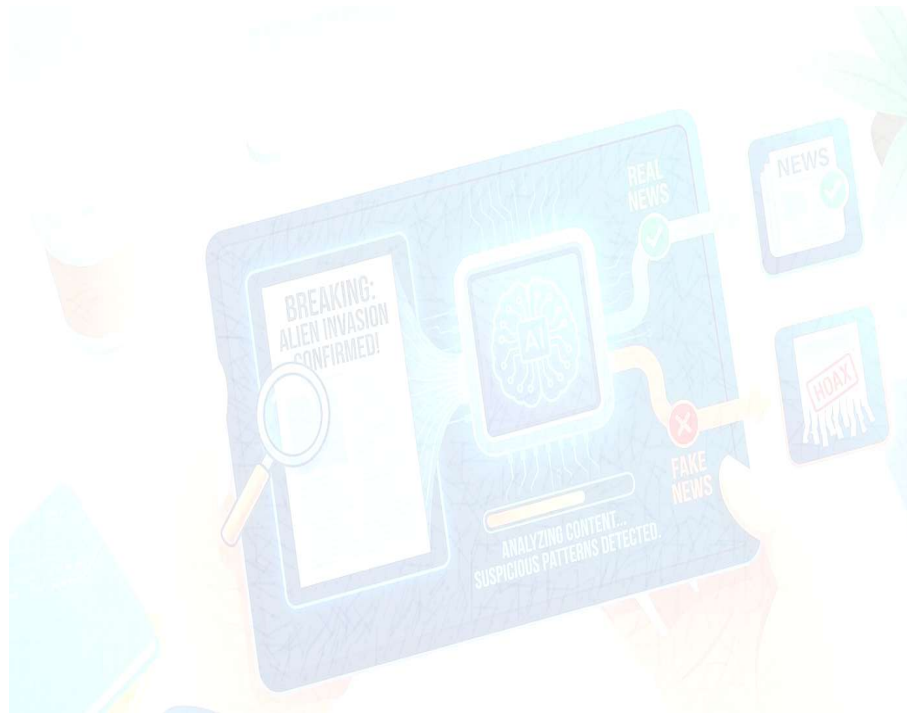
Limitations

- ▶ Depends on Dataset Quality
- ▶ Language Dependency
- ▶ Sarcasm Detection is Difficult



Future Scope

- ▶ Deep Learning Models
- ▶ Real-Time Detection
- ▶ Multi-language Support
- ▶ Browser Extension



Thank You



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